

José Hiram Soltren

1201 West Park Street, Cedar Park, TX 78613-2801, United States

✉ jsoltren@alum.mit.edu ☎ +1 (347) 503-9558 🌐 <https://www.linkedin.com/in/jsoltren>

SYNOPSIS	Pilot seeking to lead, grow, and inspire as a professional aviator. Currently self employed as an independent flight instructor. Tech industry veteran with numerous wins in prior engineering career.	
PILOT PROFILE	Commercial Pilot ASEL/AMEL #4350298 Flight Instructor CFI/CFII #4350298 Exp. 2026-10-31 Ground Instructor AIGI, Part 107 UAS First Class Medical (corrective lenses required) High Performance, Complex, High Altitude endorsements <i>No accidents, incidents, violations or deviations. Current as of February 17th, 2025</i>	
EXPERIENCE	Founder Soltren Consulting LLC, Cedar Park, TX	Jan 2016 - Present
	• Early research and design work for a certified IFR navigation device compliant with RTCA DO-160, DO-178 and DO-254. • Implemented support for a novel deep learning inference accelerator for PyTorch and ONNX backends in C++11 and Python. • Hardware and firmware development for computer keyboards in C, C++, and some Rust under “XMIT Keyboards” brand.	
	Connectivity Engineer ForeFlight	Jan 2024 - Nov 2024
	• Led third party hardware integration efforts with Tier 1 avionics platforms. • Designed and implemented API for generic third-party avionics device integration. • Avionics lab manager. Installed numerous avionics test benches.	
	Team Lead/Manager, Build and Release Roku	Jan 2023 - Sep 2023
	• Responsible for official build, release, and CI infrastructure. • Led team of four to implement self-service efforts, reducing ticket counts by 30% while improving turnaround time and throughput. • Co-led pre merge CI for Roku firmware. Presented at IEEE ICST 2023.	
	Senior Software Engineer, Roku Players Roku	Jun 2020 - Dec 2022
	• Software lead for YouTube certification for three generations of player platforms. • Software lead and hardware co-lead for Roku Streaming Stick 4K. Solved numerous hardware and software issues to unblock 20M unit program. • C++20 based firmware development with a special focus on video decode. • Authored a video capture tool widely used by external vendors and internal QA.	
	Deep Learning Software Engineer Nervana / Intel Corp.	Aug 2017 - Feb 2019
	• Driver development for “Lake Crest”. 10 MLOC multithreaded templated C++11. • Ran comparisons, benchmarks, and projections for different numerical formats on novel hardware accelerator for AI deep learning training, e.g. ResNet. • Focus on hardware and software to run GEMM and convolution as fast as possible. • Designed an algorithm for all-to-all data transfers on our hardware that beat the Bruck Algorithm, the fastest general purpose algorithm. • Wrote accelerator kernels for matrix multiplication and multi-chip communication. • Implemented support for running multiple jobs on a single chip. • Consolidated HW initialization and management for increased perf and simpler code.	

Software Engineer Cloudera, Inc.

May 2016 - Aug 2017

- Hacking on Apache Spark, a big data computation framework written in Scala.

Senior System Software Engineer NVIDIA Corporation May 2011 - Apr 2016

- Software dev on 50 MLOC multithreaded C99/C++98 graphics/OpenGL driver.
- Video decode and post-processing technical lead. Added H.265/HEVC decode support to VDPAU. Wrote stream parser for H.265/HEVC video streams.
- Fixed hundreds of bugs, including a GPU crash when playing certain videos, and a race condition in the driver when initializing multiple GPUs.
- Wrote a custom debugger to use hardware watch points to catch data races without perturbing timing.

FLYING	Commercial Pilot, Instrument Airplane, ASEL Remote Pilot, Small Unmanned Aircraft System • Co-owner of a 1978 PA-32RT-300 Piper Lance II. Single engine, high performance, complex airplane. • Manage annual inspections and equipment upgrades for the aircraft. • Participate in owner-assisted maintenance and upgrades, such as LED lighting. • 238.7 hours in type as of 2023-11-08. • 407 hours total time as of 2023-11-08. • Instrument current as of this writing. • Working toward Certificated Flight Instructor (CFI). • FAA First Class Medical.	2012 - Present
COMMUNITY	President Experimental Aircraft Association (EAA) 1607, Lago Vista, TX • Coordinate and organize chapter events. Weekend Residential Faculty Eastside College Preparatory School, East Palo Alto, CA • Operated weekend dormitory program for high school students. Big Brother Big Brothers and Big Sisters of the Bay Area, Milpitas, CA • Served as friend and mentor to a young man through his formative years. Trip Leader Bike & Build, Inc., Brooklyn, NY and Philadelphia, PA • Planned, led, and completed two cross-USA charity bicycle tours for young adults.	Jun 2023 - Present Aug 2012 - Jul 2013 Jan 2011 - Dec 2012 Jan 2006 - Aug 2008
EDUCATION	Massachusetts Institute of Technology, Cambridge, MA <i>Master of Engineering and S.B., Electrical Engineering and Computer Science</i> Grad level operating systems course. Build a kernel and hypervisor in C. Lab assistant for undergraduate microcontroller/firmware course.	
CITIZENSHIP	United States	